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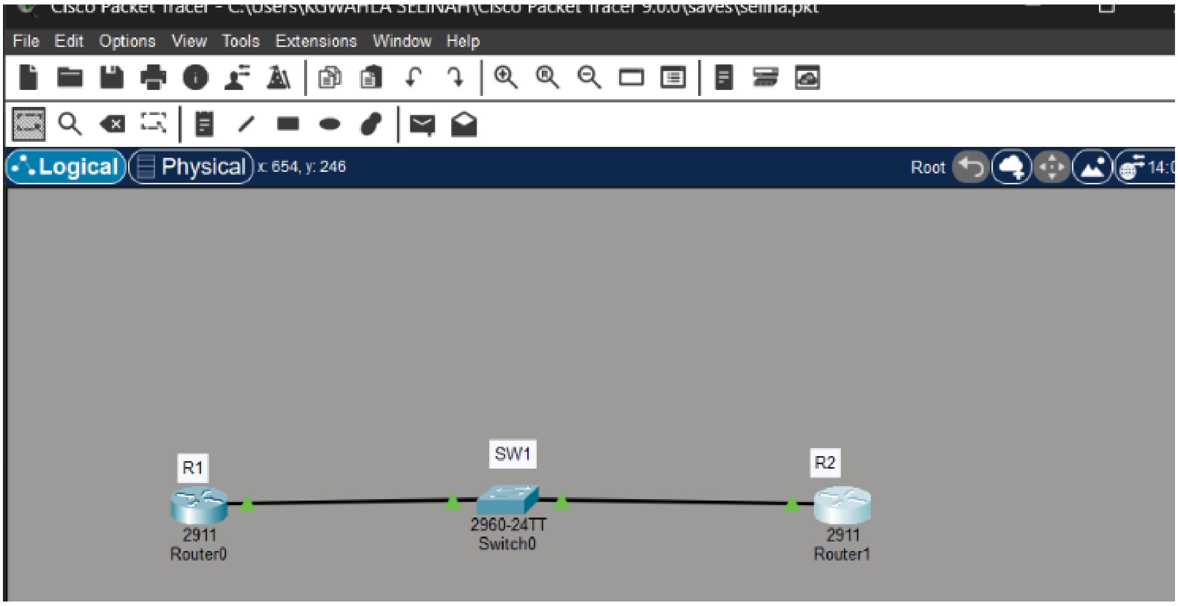
NET512 WEEK 05 PRACTICAL

- Topology diagram:



Task 32 – Task 33: IPs configured and assigned for R1, R2 and SW1

```
GigabitEthernet0/1     unassigned      YES manual down                    down
GigabitEthernet0/2     unassigned      YES manual down                    down
Vlan1                  10.10.10.10     YES manual up                      up
SW1#
SW1#
SW1#
```

```
R1>en
R1#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.1      YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

```
R2#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.2      YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

Task 34: Ping verification from the SW1

```
SW1#ping 10.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

SW1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

Task 36-39:

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int g0/0
R1(config-if)#no cdp enable
R1(config-if)#show cdp neighbors
                ^
% Invalid input detected at '^' marker.

R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce    Holdtme    Capability   Platform    Port ID
SW1          Gig 0/0          120            S        2960        Fas 0/1
R1#
```

```
R1>en
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce    Holdtme    Capability   Platform    Port ID
R1#
```

Task 40-46:

```
R2#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.2      YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

```
SW1(config-if)#duplex full
SW1(config-if)#exit
SW1(config)#int fa0/2
SW1(config-if)#speed 10
SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show interfaces fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0010.11c7.1402 (bia 0010.11c7.1402)
  Description: link to R2
 BW 100000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 10Mb/s
  input flow-control is off, output flow-control is off
```

Task 47: Reflection Question

When I set SW1's port to half duplex while R2 stayed on full duplex, the link stayed up, but it didn't work right. The connection got a lot slower, and when I checked the interface stats, I saw errors building up, like collisions and CRC errors. It also felt unstable, sometimes it worked, sometimes it dropped packets, even though the interface never showed as down.

This happens because full duplex and half duplex don't follow the same rules. Full duplex sends and receives at the same time and doesn't check for collisions. Half duplex can only do one at a time, so it keeps watching for collisions. When one side is full and the other is half, the half-duplex side thinks the other side's steady traffic is a collision, so it keeps throwing errors and resending data. That's why the link stays up but performs badly, the cable and ports are fine, the two ends just aren't talking the same way.